1) $-10°C$ Ambient temp

a) $R_{in} = 50\,\Omega$

1V peak-to-peak

$$\frac{V_p}{\sqrt{2}} = \frac{\frac{V_{pp}}{2}}{\sqrt{2}} = \frac{0.5}{\sqrt{2}}$$

$$S_{max} = \frac{V_{rms}^2}{R} = \frac{0.354^2}{50} = 2.5\,mW \sim 4\,dBm$$

b) Quantisering

$$\Delta = \frac{V_{FS}}{2^N} = \frac{1}{2^{12}} = 2.44 \cdot 10^{-4}\,V$$

$$V_{n,rms} = \frac{\Delta}{\sqrt{12}} = 7.05 \cdot 10^{-5}\,V$$

$$P_q = \frac{V_{n,rms}^2}{R} = 9.94 \cdot 10^{-11}\,V$$

$$P_q[dBm] = -70\,dBm$$

c) $f_s = 20\,MHz$ Nyquist

$$\frac{20}{2} = 10\,MHz$$

d) $N_q = -70\,dBm$

De 4:1 er termisk støj er 60 dB højere

$N_{termisk} = -60\,dBm$

$$P_n = kTB$$

termisk støj

10 MHz

$-10° = 263\,K$

+ 290,15

$= -104.4\,dBm$

$G_{tot} = 30\,dB$

$-104.4 + 30 = -74.4\,dBm$

$-74.4 + G_2 = -60\,dBm$

$G_2 = 14\,dB$

e) Brightness temp til dag vende mod

stråingsdiagram til antenne

$$T_B = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} D(\theta,\phi)\, T_B(\theta,\phi) \sin\theta \, d\theta \, d\phi$$

$$T_A = T_{Sunset} \cdot 0.75 + T_{sky} \cdot 0.25 = 274\,K$$

$$\Delta N_{out} = 10 \log_{10}\left(\frac{227K + 75.4K}{263K + 75.4K}\right) = 0.34 \text{ dB}$$

$$= 10 \log_{10}\left(\frac{P_T}{P_S}\right)$$

$$= 10 \log_{10}\left(\frac{T_S}{T_T}\right)$$

$$= 10 \log_{10}\left(\frac{T_0 + T_{ant}}{T_0 + T_{LO}}\right)$$